Arpan Saha

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Technical Skills

Languages: Java, TypeScript, JavaScript, Python, C++, SQL

Full-Stack : React, Next.js, Spring Boot, REST APIs, PostgreSQL

Cloud & DevOps: AWS Lambda, EC2, Elastic Beanstalk, RDS, S3, Docker, GitHub Actions, CI/CD, Linux

AI Engineering: LLMs, RAG, Embeddings, Semantic Search, MCP, Agentic Workflows

Testing: JUnit, Pytest, Jest, Selenium

Work Experience

Full-stack Developer — June 2026 –

ATS Containers

- Led end-to-end development of customer-facing product discovery and conversion flows, using user and market analytics to guide design and PHP/JavaScript improvements that **increased sales inquiries** by **35%**, **conversions** by **15%**, and Core Web Vitals by **30%**.
- Designed and deployed a serverless **Python agentic workflow** on **AWS Lambda** and **EventBridge** that analyzes Search Console and website data and generates recurring research insights for marketing and product decisions, with automated **Pytest** validation.
- Built an **MCP server** connecting product data and availability to LLM clients, enabling conversational product discovery and quote requests.

Production Software Developer — Oct 2025 – May 2026

Alorica

- Developed and maintained production **Java Spring Boot REST APIs** and **PostgreSQL** services, delivering application features, database migrations, and SQL query improvements that increased API reliability and performance.
- Debugged production incidents and built **GitHub Actions CI/CD** workflows automating **Maven** builds, **JUnit** tests, and deployments, improving release reliability and reducing manual deployment effort.
- Worked closely with client and support teams to understand operational issues and feature needs, then translated those requirements into backend changes, SQL scripts, and production fixes for customer care applications.

Front-End Web Developer — Apr 2022 – Apr 2026

McGill University

- Collaborated with stakeholders across multiple university departments to understand their needs and translate them into web solutions, leading the end-to-end redesign of **user journeys**, **information architecture**, and **navigation** that increased engagement by **350% (15K → 65K monthly users)**.
- Translated **Figma UI/UX designs and user journeys** into responsive, accessible web interfaces across **2,000+ webpages**, applying WCAG standards and modern frontend practices to improve usability and cross-device experiences.
- Built **Python/Selenium** tooling to automatically detect accessibility issues across webpages, helping maintain WCAG compliance across a 2,000+ page web environment.

Projects

CampusVibe | *Website* | *Source Code* — **NextJS | Java Spring Boot | PostgreSQL | AWS**

- Designed and built CampusVibe, a full-stack event discovery platform, owning the frontend, backend, database, authentication, cloud infrastructure, and deployment pipeline using **Next.js, Spring Boot, PostgreSQL**, and **AWS**.
- Designed secure **REST APIs, relational database schemas, OAuth 2.0 authentication, role-based access control, and administrative workflows** supporting students, clubs, events, bookmarks, and personalized discovery.
- Built AI-powered semantic search and personalized event planning using **LLMs**, **embeddings**, **RAG** and **user context**.
- Deployed the platform using **Elastic Beanstalk**, **RDS**, and **S3** with private database networking, IAM-based access, presigned media workflows, **Docker**, automated testing, and **GitHub Actions CI/CD**.

Education

McGill University — September 2021 – August 2025

Bachelor of Arts — Major in Computer Science and Minor in Statistics

- **Leadership Roles**: VP IT, SRI McGill; Volunteer, Care Through Tech; Captain, intramural soccer and volleyball
- **Research Projects**: Optimized a **CNN-based** medical image segmentation model in collaboration with an oncologist for **soft-tissue cancer research**; Investigated silo formation across political topics in social media using user survey info, automated data collection and statistical analysis ↗.